

Yashh Shenoy

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Summary |

Electronics and Embedded Systems Engineer with hands-on experience developing and validating embedded systems for a **Formula Student EV**, spanning **VCU/ECU control**, **CAN-based data acquisition**, **BMS development**, **hardware–software integration**, and **electric powertrain testing**. Experienced with **STM32**, **ATmega**, **ESP32**, and **NI myRIO/LabVIEW**, embedded C, multiple communication protocols, bench instrumentation, and vehicle-level fault diagnosis. B.Tech Electronics Engineering, CGPA 9.76/10, expected graduation August 2027.

Skills

Languages	C/C++ , Embedded C, Python, Java
Embedded	NI myRIO, STM32, ATmega, ESP32
Tools	LabVIEW, MATLAB/Simulink, STM32CubeIDE, STM32CubeMX, NDrive, IPG CarMaker, Xilinx Vivado
Protocols	CAN, SPI, DMX, UART, I2C, TCP, ADC, DAC, PWM
Systems	VCU, BMS, DAQ, Telemetry, Power Train

Experience

Orion Racing India – Formula Student Team, KJSCE

Feb 2024 – Aug 2026

Embedded Systems Lead & Low Voltage Electronics Engineer

Mumbai, India

- Earned **Runner-up, DigiKey Embedded Systems Innovation Award** at Formula Bharat 2026 by leading firmware for the 2026 EV's **Main VCU** on NI myRIO-1900/LabVIEW, covering 6 control functions: throttle control, traction control, regenerative braking, wireless telemetry, thermal pump control, and Ready-to-Drive sequencing.
- Improved vehicle acceleration time by **26.4% (7.2 s at Formula Bharat to 5.3 s at Formula Student Germany)** by tuning the **BAMOCAR D3** motor controller (PI control, Id, Iq, ramp functions, throttle transfer functions, field weakening) and validating powertrain integration in IPG CarMaker.
- Enabled real-time vehicle monitoring and control across **10 sensor types** (wheel speed, suspension travel, flow rate, NTC temperature, RPM, current, voltage, brake pressure, rotary position, IMU) by implementing **signal conditioning** and real-time processing in the VCU.
- Migrated core VCU functions from LabVIEW to **embedded C** on a custom **STM32H753ZI** VCU (throttle, sensor DAQ, Ready-to-Drive, thermal pump control), verified through **SIL and HIL testing** on STM32 NUCLEO boards with STM32CubeIDE.
- Unified vehicle data flow across **UART, CAN, and TCP** by building a custom CAN gateway that routes VCU sensor data to the vehicle CAN network, the **Vector data logger**, and a wireless ground-station telemetry link.
- Strengthened battery safety and monitoring by contributing to the EV **Battery Management System (BMS)**: fault detection, CAN communication, real-time monitoring, State of Charge estimation, and passive balancing, with pack simulations in **MATLAB/Simulink Battery Builder**.
- Improved vehicle reliability before competition by leading electronics debugging, testing, and validation across **low-voltage electronics, battery systems, powertrain, and embedded systems**, and supporting LV harness manufacturing, shutdown circuitry, and hardware–software interface fixes.
- Added a secondary telemetry path by developing links with **LoRa SX1276 and nRF24**, supporting full-vehicle embedded and electrical validation during Formula Student competition deployment.
- Helped the team clear scrutineering at **3 competitions** (Formula Bharat, Formula Student Austria, Formula Student Germany) by co-leading **electrical technical inspections** and assisting with **accumulator technical inspections**.

Shenoy Studio Lighting Systems

Dec 2024 – Jan 2025

Research & Development Intern

Mumbai, India

- Enabled DMX-controlled dimming of halogen lighting fixtures by leading development of an **ESP32-based DMX-to-TRIAC control system** that maps DMX input to TRIAC-based AC power control.
- Delivered manual and **DMX-controlled LED** intensity dimming and colour-temperature control by assisting in the design of **ESP32 control cards and MOSFET drivers**.
- Enabled web-based operation of curtain motors by assisting in R&D of a **web-based control system** covering embedded control, actuator interfacing, and system integration.

Education

K.J. Somaiya College of Engineering, Somaiya Vidyavihar University
Bachelor of Technology in Electronics Engineering – CGPA: 9.76/10

Aug 2023 – Aug 2027
Mumbai, India

Certifications

- Automotive Skills Development Council (India) – EV Safety Level 3E

Projects

Intelligent Performer-Tracking Stage Light

Final Year Project

- Tracked a performer in real time and steered a motorized moving-head light by fusing **4 UWB anchors plus a wearable tag, YOLO-based computer vision, and a Kalman filter** on an ESP32 with DMX512 over RS-485.
- Smoothed noisy position estimates and predicted performer motion for continuous pan/tilt control by implementing **Kalman-filter fusion** of UWB and visual measurements.
- Closed the control loop from position estimate to **pan/tilt angle calculation, DMX mapping, and fixture control**, validated on a webcam-based prototype that adjusts fixture position from tracking error.

Three-Axis Handheld Video Stabilization Gimbal

Personal

- Stabilized handheld motion within a $\pm 1^\circ$ band with **280 ms pitch and 320 ms roll settling times** (6.5% and 7.8% overshoot) by implementing **PI-D control with integral anti-windup and PWM slew-rate limiting**.
- Built a low-cost 3-axis gimbal on an **Arduino Nano, MPU6050 IMU, and 3 SG90 servos** with a custom 3D-printed frame, using **MPU6050 DMP, I2C, PWM, and C/C++** for yaw-pitch-roll estimation and servo actuation.

Pipelined RISC-V RV32IM Processor

Personal

- Built a **5-stage pipelined RV32IM core** by implementing R-, I-, S-, B-, U-, and J-type instructions plus the **RV32M** hardware multiply/divide extension, with IF/ID, ID/EX, EX/MEM, and MEM/WB pipeline registers.
- Preserved correct execution across hazards by adding **data forwarding, pipeline stalls, and branch handling**, verified through simulation and **Xilinx Vivado FPGA hardware validation**.

Interests

- **Digital Cinematography:** Certified in Professional Approach to Digital Cinematography; avid cinephile interested in filmmaking and visual storytelling.
- **Music:** Electric Guitar; Trinity College London examinations up to **Grade 6**.
- **Sports:** Track Sprinting, Football, Hiking, and Trekking.